

Lab 6 — Op Amp Circuits II: Bandpass Filter

Abdon Morales

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Description

The goal of this lab was to build a bandpass filter for the voice range, roughly 300 Hz to 3 kHz, by cascading a passive RC low-pass filter with a passive RC high-pass filter. The low-pass stage sets the upper edge of the band and the high-pass stage sets the lower edge. We built two versions of the circuit and compared them. The first version connects the two filter stages directly, with no buffering. The second version inserts a unity-gain ($\times 1$) op-amp between and after the stages so that one stage does not load the other. We swept the input frequency across the band in both LTSpice and on the trainer and recorded the peak output voltage at each frequency.

Procedure

Prelab (LTSpice).

1. We picked $R = 10\text{ k}\Omega$ for both stages and solved $f_o = 1/(2\pi RC)$ for the capacitor values that place the cutoffs at 3 kHz (low-pass) and 300 Hz (high-pass), which gives 5.3 nF and 53 nF respectively. The bench build later used the nearest stock parts, 4.7 nF and 47 nF.
2. We built the non-isolated bandpass filter in LTSpice: a voltage source into the low-pass stage, and the low-pass output wired straight into the high-pass input.
3. We set the source to a SINE function (DC offset 0, amplitude 5 V) and used the `.tran 10m` directive instead of `.op` so we could watch the transient output.
4. We stepped the source frequency through 100 Hz, 300 Hz, 500 Hz, 1 kHz, 3 kHz, 5 kHz, 10 kHz, and 100 kHz, recording the peak output voltage each time.
5. We built the isolated version using the same components plus two unity-gain op-amps. The low-pass output drives the non-inverting input of the first op-amp, whose output is tied back to its own inverting input and also feeds the high-pass input. A second unity-gain op-amp buffers the high-pass output. We added `.inc opamp.sub` and reran the same frequency sweep.

In-lab (XK700 trainer).

1. We verified each resistor and capacitor value before wiring anything.
2. *Part 1:* We built the low-pass and high-pass stages on the trainer with the low-pass output feeding the high-pass input, driven by the trainer's sine generator. Using the oscilloscope we set the input to 5 V peak-to-peak, put the low-pass input on CH1 and the high-pass output on CH2, and swept the frequency from 100 Hz to 100 kHz, recording the output at each point.
3. *Part 2:* With the power off, we added the two op-amps as unity-gain buffers, powered them from the +12 V / -12 V supply rails, rechecked the wiring, and repeated the same sweep, measuring at the output of the second op-amp.
4. We noted any unusual behavior in both builds.

Simulations

We ran two LTSpice simulations: the non-isolated (passive) cascade and the isolated version with two unity-gain op-amp buffers. In both cases the source is a SINE function (DC offset 0, amplitude 5 V) and we used `.tran 10m` so we could read the peak output off the transient waveform. The frequency was changed for each run.

Non-isolated (passive) bandpass filter

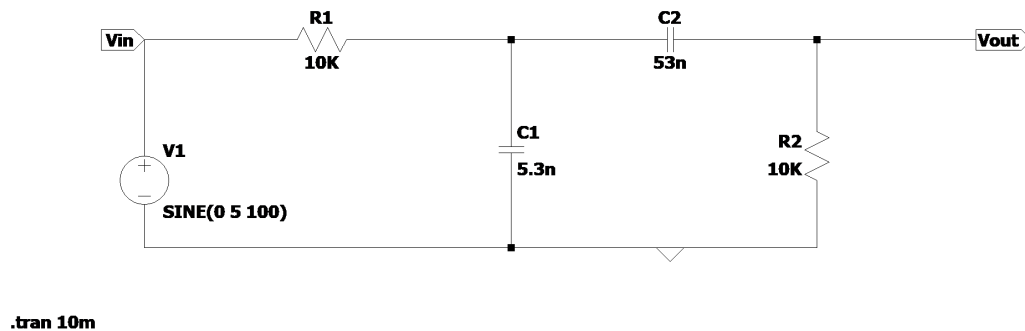


Figure 1: Passive bandpass filter: low-pass stage cascaded directly into the high-pass stage.

```
* /Users/abdonmorales/Library/CloudStorage/OneDrive -  
  TheUniversityofTexasatAustin/Summer 2026/ECE 402/Lab Notes/Lab 06/lab06  
  -part1.asc  
* Generated by LTspice 26.0.2 for MacOS.  
V1 Vin 0 SINE(0 5 100)  
R1 N001 Vin 10K  
R2 0 Vout 10K  
C1 N001 0 5.3n  
C2 Vout N001 53n  
.tran 10m  
.backanno  
.end
```

Listing 1: LTSpice netlist: passive cascade

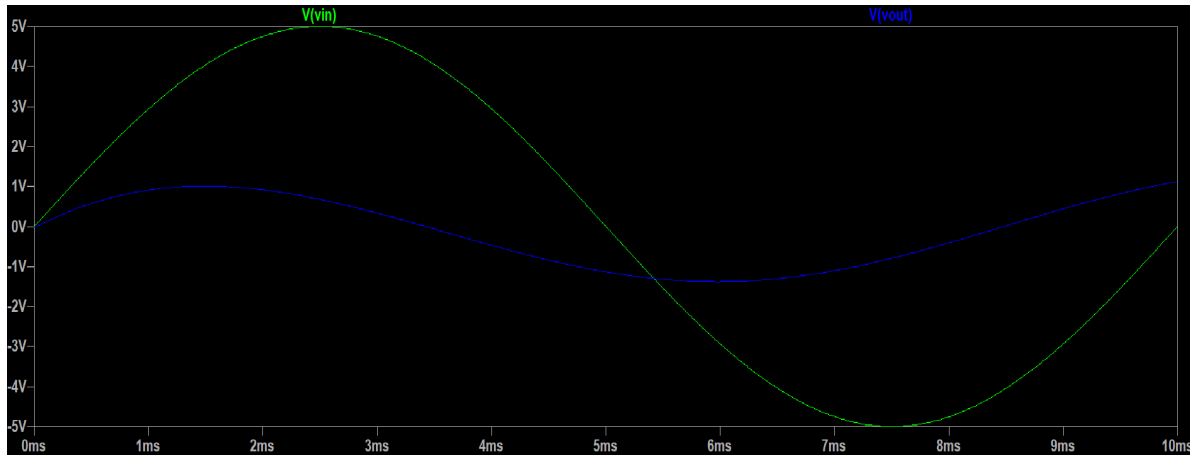


Figure 2: Input signal (green) and high-pass output (blue) for the passive cascade.

Isolated (op-amp buffered) bandpass filter

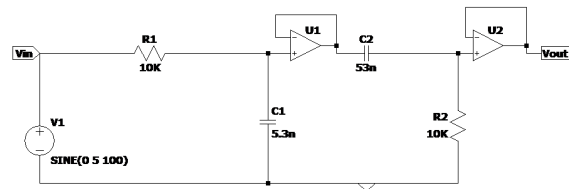


Figure 3: Buffered bandpass filter: unity-gain op-amps isolate the two stages.

```
* /Users/abdonmorales/Library/CloudStorage/OneDrive -
  TheUniversityofTexasatAustin/Summer 2026/ECE 402/Lab Notes/Lab 06/lab06
  -part2.asc
* Generated by LTspice 26.0.2 for MacOS.
V1 Vin 0 SINE(0 5 100)
R1 N002 Vin 10K
R2 0 N003 10K
C1 N002 0 5.3n
C2 N003 N001 53n
U1 N001 N002 N001 opamp Aol=100K GBW=10Meg
U2 Vout N003 Vout opamp Aol=100K GBW=10Meg
.tran 10m
.inc opamp.sub
.backanno
.end
```

Listing 2: LTSpice netlist: op-amp buffered version

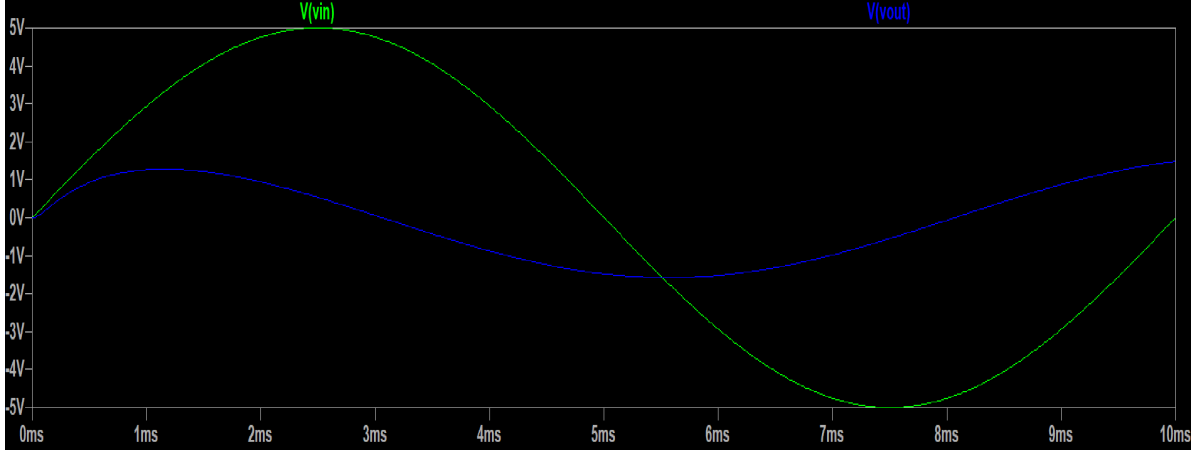


Figure 4: Input signal (green) and output (blue) for the buffered version.

The peak output from each run is tabulated next to the measured bench data in the Measurements section for direct comparison. These simulations use the calculated caps (5.3 nF and 53 nF) shown in the netlists above, while the bench used the stock 4.7 nF and 47 nF parts.

Measurements

The peak output voltage at each test frequency is below for both builds, with the LTSpice value and the bench value side by side. The simulation used the calculated caps (5.3 nF and 53 nF); the bench build used the stock parts (4.7 nF and 47 nF). The last column is the percent difference of the measured value relative to the simulation.

Table 1: Circuit 1: passive cascade (no op-amps), $V_{in} = 5\text{ V}$

Frequency	V_{out} simulated (V)	V_{out} measured (V)	Diff. (%)
100 Hz	1.101	1.24	+12.6
300 Hz	2.164	2.08	-3.9
500 Hz	2.305	2.26	-2.0
1 kHz	2.340	2.32	-0.9
3 kHz	2.153	2.16	+0.3
5 kHz	1.820	1.90	+4.4
10 kHz	1.209	1.40	+15.8
100 kHz	0.136	0.32	+135.6

Table 2: Circuit 2: buffered version (unity-gain op-amps), $V_{in} = 5\text{ V}$

Frequency	V_{out} simulated (V)	V_{out} measured (V)	Diff. (%)
100 Hz	1.452	1.34	-7.7
300 Hz	3.500	3.14	-10.3
500 Hz	4.184	3.54	-15.4
1 kHz	4.447	3.76	-15.4
3 kHz	3.500	3.30	-5.7
5 kHz	2.446	2.54	+3.8
10 kHz	1.333	1.44	+8.0
100 kHz	0.135	0.24	+77.8

Both circuits peak at 1 kHz, near the geometric center of the band, which is what a bandpass response should do. The passive cascade peaks at about 2.3 V while the buffered version reaches about 3.8 V on the bench (4.4 V in simulation), so adding the op-amps roughly doubles the passband output.

Observations & Discussion

There was a clear difference between the two circuits, and the cause is loading. In the passive cascade the high-pass stage draws current directly from the low-pass output, so the first stage is loaded down and the passband peaks at only about 2.3 V. Adding the unity-gain op-amps isolates the stages and approximately doubles this output, reaching about 3.8 V on the bench and 4.4 V in simulation, because each buffer presents a high input impedance to the preceding stage and a low output impedance to the following one. Both versions peak at 1 kHz, the center of the band, which is the expected behavior for a bandpass response. Which circuit matched expectation depends on the reference used: the passive build agreed closely with its own simulation, within about five percent from 300 Hz to 5 kHz, while the buffered build reproduced the ideal textbook bandpass shape more faithfully (a taller, cleaner peak) but fell about ten to fifteen percent below the ideal simulation across the passband. This shortfall represents the primary cost of the op-amps and is attributable to the real 741 limitations (finite gain-bandwidth, input bias, and offset) that the LTSpice model does not capture, in addition to the two extra components and the $\pm 12\text{ V}$ rails they require. For a voice-band filter the tradeoff is justified, since the passive version discards a substantial portion of the in-band signal.

The main concern appeared at 100 kHz, where both bench readings (0.32 V passive and 0.24 V buffered) remained well above the simulated floor of about 0.14 V. The physical breadboard does not attenuate out-of-band signals as effectively as the ideal model, which we attribute to high-frequency feedthrough through parasitic capacitance in the board and probes. The buffered measurements also fell uniformly below simulation across the passband, whereas the passive measurements straddled it. This offset reflects error introduced by the op-amps, which the purely passive circuit does not have. The capacitor substitution shifted the band as well. The design called for 5.31 nF and 53.05 nF, which is what we simulated, but these are not standard values, so the bench build used 4.7 nF and 47 nF. Smaller capacitance raises the cutoff frequency, so both edges moved up: the low-pass corner went from 3 kHz to about 3.39 kHz and the high-pass corner from 300 Hz to about 339 Hz, which widened the passband slightly and shifted it up by approximately thirteen percent on each edge. This component-value difference, together with resistor and capacitor tolerances, partly explains the gap between the measured and simulated values, though for voice filtering the shift is negligible.